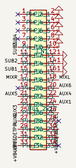
POWER ENTRY, LOGIC SUPPLY, GROUNDS, DECOUPLING

R1 10R
C1 220u 25V
C2 100n
R2 10R
C3 220u 25V
C4 100n
R3 10R
C5 47u 25V
R4 10R
C6 47u 25V
NT1 AGND–LGND star
CN4 Ground spade 6.3mm
R5 10R GND cancel
U1C OPA1612
C20 100n
C21 100n
U2C OPA1612
C22 100n
C23 100n
U3C OPA1612
C24 100n
C25 100n
U5C OPA1612
C29 100n
C30 100n
U6C OPA1612
C31 100n
C32 100n
U7C OPA1612
C33 100n
C34 100n
U8C OPA1612
C35 100n
C36 100n
U4C LM393
C26 100n
#FLG01 PWR_FLAG
#FLG02 PWR_FLAG
#FLG03 PWR_FLAG
#FLG04 PWR_FLAG
#FLG05 PWR_FLAG
#FLG06 PWR_FLAG

MOTHERLOOM (Delta pinout; one header, crimped socket on a daisy-chained ribbon)



LINE INPUT — ground-compensated receiver (inverting), Ø switch SW15: latched = phase reversed

CN3 Jack board link 2x5
R10 1M
SW15A Phase
SW15B Phase
R6 FDK 0.1%
R7 10K 0.1%
C7 33p COG
R8 FDK 0.1%
R9 10K 0.1%
U1A OPA1612

GAIN −15..+15 dB, 0 dB at the detent (inverting; pot pin 3 = CW = input end)

R45 2K2 1%
RV9 Gain 10K lin CD
R46 2K2 1%
C37 22p COG
U1B OPA1612
C38 47u NP 25V
R47 47K

HPF unity Sallen-Key Butterworth, −3 dB 87 Hz SW11: latched = HPF in (unlatched shorts the caps)

C39 47n COG
C40 47n COG
R48 27K
R49 56K
U5A OPA1612
SW11A HPF
SW11B HPF

INSERT POSITION JUMPERS 1–2 = insert post-EQ (default); 2–3 on all three = insert pre-EQ

JP6 EQ input
JP5 Insert send source
JP7 PFL source

EQ Delta ED3344A/ED3151 values: HF shelf (U5B), HI-MID (U6A), LO-MID (U6B), LF (U7A); CW = boost

R50 33K
R51 33K
C41 33p COG
U5B OPA1612
RV10 HF 10K lin CD
C42 6n7 COG
R52 5K6
C43 6n7 COG
R53 6K8
R54 22K
R55 22K
U6A OPA1612
RV11 HI-MID gain 10K lin CD
C44 4n3 COG
R56 6K8
RV12 HI-MID freq 100K C dual
R57 6K8
C45 1n5 COG
R58 22K
R59 22K
U6B OPA1612
RV13 LO-MID gain 10K lin CD
C46 15n COG
R60 6K8
RV14 LO-MID freq 100K C dual
R61 6K8
C47 6n8 COG
R66 22K
R67 22K
U7A OPA1612
R62 3K3
R63 3K3
RV15 LF 10K lin CD
C48 150n film
R64 1K5
C49 2.2u film
C50 100n COG
R65 5K1

EQ IN SW12: latched = EQ in (unlatched = bypass from the EQ input); green EQ LED

SW12A EQ
SW12B EQ
R69 4K7
D5 EQ green
C51 47n NP 25V
R68 47K

INSERT (send = tip, return = ring, ring normalled to send) + return buffer

R11 75R
C10 47u NP 25V
R12 47K
U7B OPA1612
C11 220u NP 25V
R13 47K

OVERLOAD DETECTOR (-+18 dBu)

R39 1K
D1 1N4148
C12 1u
R14 470K
R40 10K
R41 10K
U4A LM393
R15 1K
D2 OL red
U4B LM393

PFL (pre-fade) SW3: A–B = ON

R32 22K
SW3A PFL
SW3B PFL
R33 100K
R34 4K7
D3 PFL red

ON / SOFT MUTE (2x J112 shunt, ~20 ms ramp) SW4: latched = channel ON, green LED

R16 4K7
Q1 J112
R17 4K7
Q2 J112
U3B OPA1612
SW4A ON
R42 220K
C28 100n
SW4B ON
R43 4K7
D4 ON green

FADER + POST-AMP (+10 dB; omit R38 for unity) JP1 = VCA bypass

RV1 Fader 10K log 100mm
C13 2.2u
JP1 VCA bypass
R18 100K
J3 IC socket
J4 VCA control in
R44 100K
U2A OPA1612
R19 4K7
C14 100p COG
R38 2K2
C15 220u NP 25V
R35 47K

AUX SENDS PRE switch per pair (latched = pre); jumpers 1–2 = pre-fader, 2–3 = pre-EQ. SW13 DIR: aux 6 pot -> direct out

JP2 Aux1/2 pre source
SW8A PRE 1–2
SW8B PRE 1–2
JP3 Aux3/4 pre source
SW9A PRE 3–4
SW9B PRE 3–4
JP4 Aux5/6 pre source
SW10A PRE 5–6
SW10B PRE 5–6
RV2 Aux1 10K log
R20 22K
RV3 Aux2 10K log
R21 22K
RV4 Aux3 10K log
R22 22K
RV5 Aux4 10K log
R23 22K
RV6 Aux5 10K log
R24 22K
RV7 Aux6 10K log
R25 22K
SW13A DIR
SW13B DIR

DIRECT OUT (level = aux 6 pot when DIR is latched; impedance-balanced TRS)

R70 100K
U8A OPA1612
R71 75R
C52 47u NP 25V
R72 47K
R73 75R
U8B OPA1612

PAN + PAN BUFFERS (AC-coupled outputs)

RV8 Pan 10K dual
U2B OPA1612
U3A OPA1612
C16 220u NP 25V
R36 47K
C17 220u NP 25V
R37 47K

PAN SWITCH SW14: latched = subgroups panned; unlatched = subgroups get POST, unpanned (MIX always panned)

SW14A PAN
SW14B PAN

ROUTING MATRIX (A–B = assigned; off grounds the resistor) 4K7 summing

R26 4K7 1%
R27 4K7 1%
SW5A Assign L–R
SW5B Assign L–R
R28 4K7 1%
R29 4K7 1%
SW6A Assign 1–2
SW6B Assign 1–2
R30 4K7 1%
R31 4K7 1%
SW7A Assign 3–4
SW7B Assign 3–4

Nets joined by labels: tidy wiring in Eeschema as needed
Generated from channel_strip_schematic_spec.md v4.0

Sheet: /
File: channel_strip.kicad_sch

Title: Channel strip — goa console (Delta Theatre input without the mic stage)

Size: User | Date: 2026-10-19 | Rev: 4.0
KiCad E.D.A. 10.0.7 | Id: 1/1